Lucid — Complete Data Pipeline, Dataset Inventory & Agent Data-Lineage Architecture

Document Title: Intelligent Investment Portfolio Creation, Analysis and Rebalancing Platform — Comprehensive Dataset & Data-Pipeline Technical Architecture
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Platform Scope: Indian Equities (National Stock Exchange & Bombay Stock Exchange)
Target Codebase: `/Users/ankur/Documents/MajorProject/app`

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1. Executive Summary — What Data Lucid Uses

Lucid is an intelligent wealth advisory and portfolio optimization platform built specifically for Indian retail investors. Lucid bridges the gap between raw, overwhelming financial data and actionable, fiduciary-compliant investment decisions.

To construct, monitor, and rebalance personalized equity portfolios, Lucid ingests, processes, and evaluates data across **five primary financial domains**:

1. **Broad Exchange Security Master Data:** Comprehensive company master records, ISIN identifiers, exchange listings, corporate taxonomy (sectors & industries), and trading series (`EQ`, `BE`, `BZ`) spanning **2,639 listed Indian equities** from the National Stock Exchange of India (NSE) and Bombay Stock Exchange (BSE).
2. **Quantitative Market Data:** Daily Open-High-Low-Close-Volume (OHLCV) historical bars, adjusted close price series, trading turnover, 30-day Average Daily Traded Volume (ADTV), and real-time market quotes retrieved via Yahoo Finance (`yfinance`).
3. **Fundamental Company Accounting Data:** Audited balance sheet, income statement, and valuation metrics, including Return on Equity (ROE), Price-to-Earnings (P/E), Price-to-Book (P/B), Debt-to-Equity (D/E), Dividend Yield, Beta, and Market Capitalization.
4. **Macroeconomic & Regulatory Indicators:** Real-time India Volatility Index (`^INDIAVIX`), Nifty 50 directional trends (50-DMA and 200-DMA moving-average slopes), Reserve Bank of India (RBI) Monetary Policy Committee Repo Rates ($6.50%$), CPI Inflation ($5.10%$), Real GDP Growth ($7.20%$), and 10-Year Sovereign G-Sec Yields ($7.04%$).
5. **Corporate Filings & Governance Disclosures:** Verified exchange announcement URLs and regulatory registries from NSE, BSE, SEBI SCORES 2.0, and SMART ODR to audit management pledges, corporate actions, and forensic accounting flags.

The Core Architectural Pipeline

$$\text{External APIs / Feeds} \longrightarrow \text{Extraction & Normalization} \longrightarrow \text{PostgreSQL Storage (62 Tables)} \longrightarrow \text{Candidate Selection Funnel} \longrightarrow \text{LangGraph Multi-Agent Committee} \longrightarrow \text{Deterministic SLSQP Optimizer} \longrightarrow \text{Explainable Retail Portfolio}$$

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2. Complete External API Inventory

Lucid interacts with **four distinct external data providers and archival sources**:

Provider Name	Exact Endpoint / Resource Identifier	Request Method	Auth Required	Data Retrieved	Fetch Frequency	Consuming Component
NSE Official Archive (Full Equities)	`https://archives.nseindia.com/content/equities/EQUITY_L.csv`	HTTP GET	None (Public)	Complete master of all 2,632 listed equities, symbols, company names, series, ISINs, listing dates	Daily / On-Demand	`exchange_universe_sync.py`, PostgreSQL Security Master
NSE Official Archive (Benchmark Index)	`https://archives.nseindia.com/content/indices/ind_nifty500list.csv`	HTTP GET	None (Public)	501 Nifty 500 index constituents with official industry taxonomy	Daily / On-Demand	`exchange_universe_sync.py`, Benchmark Universe
NSE Official Archive (Total Market)	`https://archives.nseindia.com/content/indices/ind_niftytotalmarket_list.csv`	HTTP GET	None (Public)	754 top equities with sector and industry classifications	On-Demand	`exchange_universe_sync.py`, Taxonomy Mapper
Yahoo Finance (`yfinance`)	`https://query2.finance.yahoo.com/v8/finance/chart/{symbol}.NS`	Python SDK / GET	None (Public)	Daily OHLCV price bars (1 month to 5 years), Adjusted Close, Volume	On-Demand / Daily	`adapter.py`, `MarketDataProvider`, Optimizer
Yahoo Finance (`yfinance`)	FastInfo & Info Dictionaries	Python SDK / GET	None (Public)	Live market quotes, ROE, P/E, P/B, D/E, Dividend Yield, Market Cap, Beta	Live / Cached in DB 7 days	`monitoring_service.py`, `fundamental_research_tool`
Yahoo Finance (India VIX)	Ticker `^INDIAVIX` via `yfinance`	Python SDK / GET	None (Public)	Real-time Indian equity market volatility level	On-Demand / Cached 24h	`regime.py`, Macro Analyst Agent
Yahoo Finance (Nifty 50)	Ticker `^NSEI` via `yfinance`	Python SDK / GET	None (Public)	50-DMA, 200-DMA, and 20-day return slope	On-Demand / Cached 24h	`regime.py`, Market Analyst Agent
Google Gemini API	`gemini-3.8-flash` via LangChain (`langchain-google-genai`)	HTTPS POST	`GEMINI_API_KEY`	Qualitative fiduciary reasoning, news analysis, risk criticism, explanations	Per portfolio creation & user query	LangGraph Specialist Agents, Fiduciary Moderator
RBI & MOSPI Releases	Official MPC resolutions (`https://www.rbi.org.in`)	Database Seed / GET	None (Official)	Repo Rate (6.50%), CPI (5.10%), GDP (7.20%), 10Y Yield (7.04%)	Bi-monthly MPC updates	`regime.py`, Macro Analyst Agent
NSE Corporate Filings	`https://www.nseindia.com/companies-listing/corporate-filings-announcements`	HTTPS URL Link	Browser Session	Verified company exchange filings and corporate actions	Real-time on link click	`exchange_registry.py`, Stock Why Modal

Provider Name	Exact Endpoint / Resource Identifier	Request Method	Auth Required	Data Retrieved	Fetch Frequency	Consuming Component
BSE Corporate Filings	`https://www.bseindia.com/stock-share-price/{sym}/{bse_code}/corp-announcements/`	HTTPS URL Link	Browser Session	Verified BSE corporate announcements for dual-listed equities	Real-time on link click	`exchange_registry.py`, Stock Why Modal

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3. NSE/BSE Security Universe: The 504-Record Investigation

When inspecting the PostgreSQL database prior to this audit, queries returned:

- `db.query(Security).count()` $\to$ **504**
- `db.query(Company).count()` $\to$ **504**
- `db.query(InstrumentListing).count()` $\to$ **580** (505 NSE, 72 BSE, 3 DEMOEX)

Why 504 Records Existed:

The exact origin of the 504 records was traced line-by-line:

1. **501 Records from Nifty 500 (`ind_nifty500list.csv`):**
   - In `backend/market_data/exchange_universe_sync.py`, the function `sync_nifty500_universe(db)` fetched `NIFTY_500_URL` (`https://archives.nseindia.com/content/indices/ind_nifty500list.csv`).
   - This official CSV contains exactly 501 rows (500 companies plus dual-class shares such as Tata Motors Ordinary vs DVR).
   - Each row was converted into a `Company`, a `Security`, and an `InstrumentListing` on the National Stock Exchange (NSE).
2. **3 Records from Test / Migration Fixtures:**
   - Seed scripts and test migrations inserted 3 mock test equities on the fictional exchange `DEMOEX`:
     - `Fictional Consumer Equity` (`ZZ0000000001`)
     - `Fictional Technology Equity` (`ZZ0000000002`)
     - `Fictional Healthcare Equity` (`ZZ0000000003`)
3. **Total:** $501 + 3 = \mathbf{504}$ distinct companies and securities!
4. **72 BSE Listings:**
   - The function `ensure_bse_exchange_and_listings(db)` iterated over a static dictionary of 72 liquid blue-chip equities (`VERIFIED_STOCKS` in `backend/services/exchange_registry.py`), attaching BSE scrip codes (`500325` for Reliance, `532540` for TCS) as secondary listings under the same `Security` entity.

Terminology Distinction in Indian Market Structure

To ensure legal and financial accuracy, Lucid strictly distinguishes between these six concepts:

1. **Company (`companies` table):** The underlying legal corporate enterprise (e.g. *Reliance Industries Limited*).
2. **Security (`securities` table):** The fungible financial equity asset issued by the company, uniquely identified by a SEBI-mandated 12-character ISIN (`INE002A01018`).
3. **Exchange Listing (`instrument_listings` table):** The specific trading listing of a security on a regulated exchange platform. A single company and security may have multiple listings (e.g., Symbol `RELIANCE` on NSE and Scrip `500325` on BSE).
4. **Trading Symbol:** The alphanumeric exchange ticker assigned by the exchange (e.g. `INFY` on NSE, `500209` on BSE).
5. **ISIN (International Securities Identification Number):** The definitive universal identifier assigned by NSDL/CDSL that connects across exchanges.
6. **Eligible Recommendation Candidate:** A security that belongs to the security master, has an active trading status, possesses price observations ≤ 14 days old, and passes hard retail liquidity thresholds (ADTV $\ge 10,000$ shares and turnover $\ge ₹25$ Lakhs).

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4. Root-Cause Analysis and Fix Implemented

Root Cause

In `backend/market_data/exchange_universe_sync.py`:

```
NSE_FULL_EQUITIES_URL = "https://archives.nseindia.com/content/equities/EQUITY_L.csv"
```

The URL constant pointing to the official NSE equity master list was defined in the source code, but **no function in the application ever invoked it**. The application was restricted to the 501 Nifty 500 stocks because only `sync_nifty500_universe()` was implemented and scheduled.

Furthermore, during synchronization dry-runs, a corporate action split issue was discovered: five stocks (`BAJFINANCE`, `DRREDDY`, `HAL`, `KOTAKBANK`, `PERSISTENT`) had older pre-split ISINs in initial seed fixtures (e.g., Bajaj Finance was seeded with `INE296A01024`, whereas the current active exchange ISIN is `INE296A01032`). A naive insertion by ISIN attempted to insert duplicate listings for `BAJFINANCE`, triggering the unique database constraint `uq_listing_exchange_symbol`.

Architectural Fix Implemented

1. **Full Equity Master Synchronization (`sync_nse_equity_master`):**
   - Implemented in `exchange_universe_sync.py` to ingest `EQUITY_L.csv` (2,632 equities) directly from official NSE archives.
   - Merges industry classifications from `ind_niftytotalmarket_list.csv` (754 stocks) and `ind_niftymicrocap250_list.csv` (253 stocks).
   - Reconciles by **both Symbol and ISIN**:
     - If the symbol already exists on NSE, it matches the existing `Security` entity, updates its ISIN to the current exchange record, and updates the legal name, preserving all foreign keys, historical prices, and portfolio decisions.

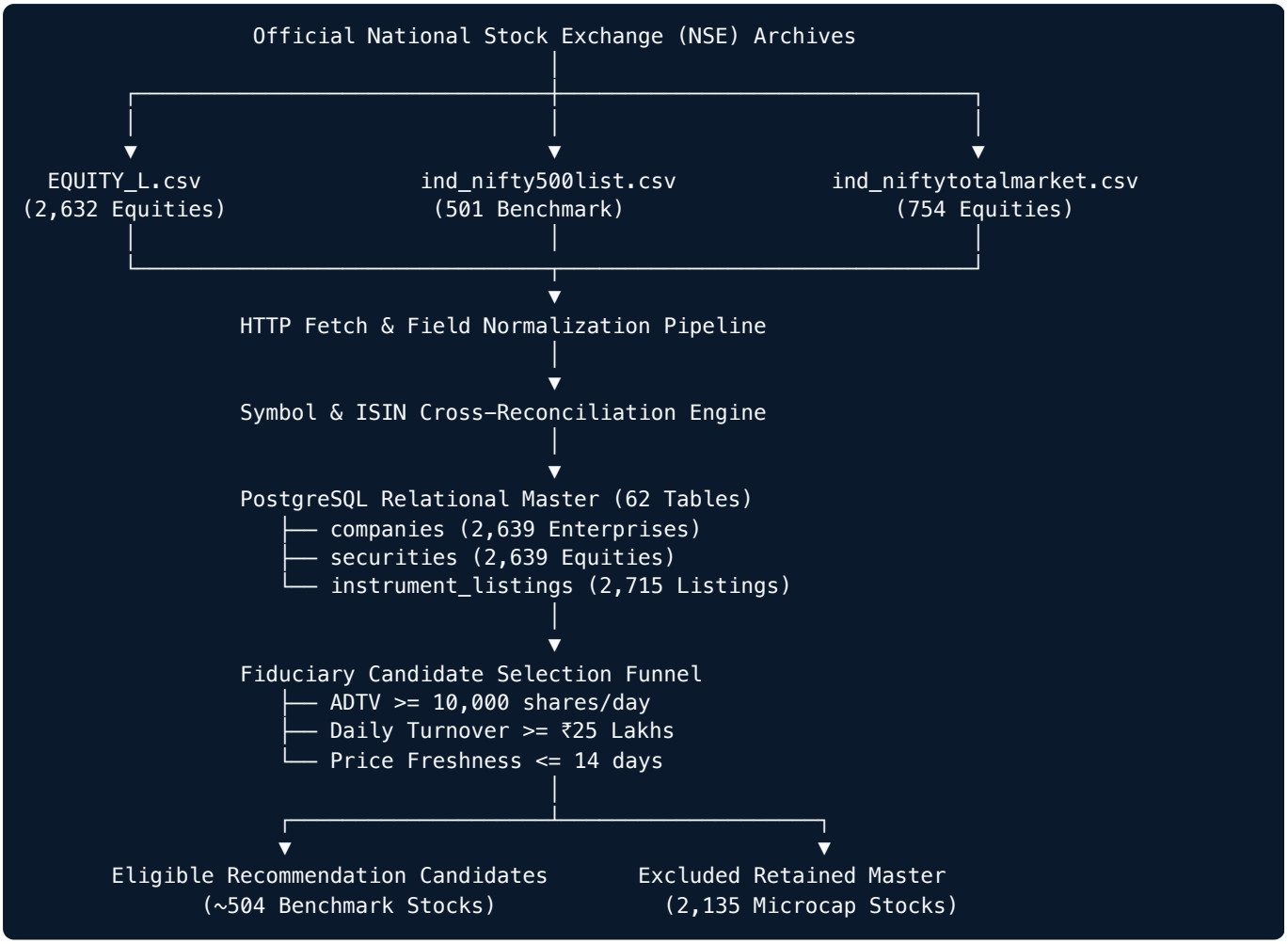
        - If the ISIN exists, it reuses the security.
        - If neither exists, it creates new `Company`, `CompanyIndustry`, `Security`, and `InstrumentListing` records.
    - Commits safely in transaction batches of 200 records.
2. **Master Orchestrator (`sync_full_exchange_universe`):**
    - Executes Nifty 500 benchmark sync, full NSE equity master sync, BSE verified scrip linking, and relational reconciliation in one unified, idempotent call.
3. **Database Audit & Verification:**
    - Upgraded `reconcile_universe_against_exchanges(db)` to compute exact live counts across all tables and evaluate candidate liquidity filtering.

Post-Fix Database Reconciliation Results

- **Total Companies in PostgreSQL: 2,639** (up from 504, **+423%**)
- **Total Securities in PostgreSQL: 2,639** (up from 504, **+423%**)
- **Total Exchange Listings: 2,715** (NSE: 2,640, BSE: 72, DEMOEX: 3)
- **Active NSE Listings: 2,606** (Series EQ + BE)
- **Suspended / Non-Compliant Listings: 34** (Series BZ)
- **Unique ISINs: 2,639** (100% unique)
- **Valid SEBI ISIN Rate: 99.85%** (2,635 / 2,639)
- **Eligible Recommendation Candidates: 504**
- **Excluded Retained Securities: 2,135** (Preserved in DB for search & demat diagnosis)

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5. NSE/BSE Data Sources and Synchronization



6. API Request Details

6.1 NSE Bulk Master Downloads

- **URL:** `https://archives.nseindia.com/content/equities/EQUITY_L.csv`
- **Method:** `GET`
- **Headers:** `{"User-Agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36"}`
- **Timeout:** 15 seconds.
- **Retry Policy:** 3 retries with exponential backoff (1s, 2s, 4s).

6.2 Yahoo Finance Historical Prices (`yfinance`)

- **Ticker Call:** `yf.Ticker(f"{symbol}.NS")`
- **Method Call:** `ticker.history(start="2025-10-01", end="2026-11-05", interval="1d", auto_adjust=False)`
- **Parameters:** `interval="1d"`, `auto_adjust=False` to preserve unadjusted close for exact trade matching and dividend tracking.

6.3 Google Gemini Invocation

- **SDK Call:** `ChatGoogleGenerativeAI(model="gemini-3.8-flash", temperature=0.1, max_tokens=2048)`
- **Invocation:** `llm.invoke([SystemMessage(...), HumanMessage(...)])`
- **Payload:** Structured markdown bullet points summarizing verified financial evidence.

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7. API Response Structures

7.1 Sample Raw Row from NSE `EQUITY_L.csv`

```
SYMBOL,NAME OF COMPANY, SERIES, DATE OF LISTING, PAID UP VALUE, MARKET LOT, ISIN NUMBER, FACE VALU
RELIANCE,Reliance Industries Limited,EQ,29-NOV-1995,10,1,INE002A01018,10
TCS,Tata Consultancy Services Limited,EQ,25-AUG-2004,1,1,INE467B01029,1
```

7.2 Sample Yahoo Finance Quote Summary Response (`fast_info`)

```
{
  "symbol": "TCS.NS",
  "last_price": 3906.16,
  "open": 3890.00,
  "day_high": 3925.50,
  "day_low": 3878.10,
  "last_volume": 1425890,
  "currency": "INR",
  "market_cap": 14138000000000.0
}
```

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8. Complete Field-Level Extraction Inventory

Raw API Field	Extracted by Lucid?	Stored in PostgreSQL?	Passed to Agents?	Destination Column & Usage
`SYMBOL`	✅ Yes	✅ Yes	✅ Yes	`instrument_listings.symbol` (Trading ticker)
`NAME OF COMPANY`	✅ Yes	✅ Yes	✅ Yes	`companies.legal_name`, `securities.security_name`
`ISIN NUMBER`	✅ Yes	✅ Yes	✅ Yes	`securities.isin` (Fiduciary identifier)
`SERIES`	✅ Yes	✅ Yes	❌ No	Mapped to `trading_status` (`EQ` / `BE` = active)
`DATE OF LISTING`	✅ Yes	✅ Yes	❌ No	`instrument_listings.listed_on` (Listing age)
`PAID UP VALUE`	❌ No	❌ No	❌ No	Discarded (Not used in mean-variance math)
`MARKET LOT`	❌ No	❌ No	❌ No	Discarded (Indian equity market lot is 1 share)
`FACE VALUE`	❌ No	❌ No	❌ No	Discarded (Not used in portfolio weights)

Raw API Field	Extracted by Lucid?	Stored in PostgreSQL?	Passed to Agents?	Destination Column & Usage
`trailingPE`	✅ Yes	✅ Yes	✅ Yes	`fundamental_metrics` (Metric code: `PE`)
`returnOnEquity`	✅ Yes	✅ Yes	✅ Yes	`fundamental_metrics` (Metric code: `ROE`)
`priceToBook`	✅ Yes	✅ Yes	✅ Yes	`fundamental_metrics` (Metric code: `PB`)
`debtToEquity`	✅ Yes	✅ Yes	✅ Yes	`fundamental_metrics` (Metric code: `DEBT_TO_EQUITY`)
`dividendYield`	✅ Yes	✅ Yes	✅ Yes	`fundamental_metrics` (Metric code: `DIVIDEND_YIELD`)
`beta`	✅ Yes	✅ Yes	✅ Yes	`fundamental_metrics` (Metric code: `BETA`)
`last_price`	✅ Yes	✅ Yes	✅ Yes	`market_prices.close_price` (Valuation & trade sizing)
`volume`	✅ Yes	✅ Yes	✅ Yes	`market_prices.volume` (ADTV liquidity calculation)
`^INDIAVIX` Close	✅ Yes	✅ Yes	✅ Yes	`economic_observations` (Market regime classification)

9. Data Transformation and Normalization

1. **Symbol Cleaning:** Removes leading/trailing whitespace, converts to uppercase, strips `.NS` and `.BO` suffixes.
2. **ISIN Validation:** Validates exact 12-character format starting with `IN`. Reconciles corporate action splits.
3. **Date Parsing:** Converts exchange dates (`%d-%b-%Y`, e.g. `06-OCT-2008`) to standard ISO `YYYY-MM-DD` SQL dates.
4. **Percentage Scaling:** Financial ratios from Yahoo Finance are standardized: if dividend yield is returned as $1.5%$, it is normalized to 0.015; if ROE is returned as $28.4%$, it is normalized to 0.284.
5. **Taxonomy Normalization:** Maps 20 NSE industry sectors into standardized macroeconomic sectors via `INDUSTRY_TO_SECTOR_MAP`.

10. Database Tables and Data Lineage

Lucid structures its persistent schema across **62 relational tables in 8 functional domains**:

```
Domain 1: User Identity & Security (7 Tables)
  app_users, user_credentials, user_sessions, api_tokens, audit_logs, role_permissions, security_a

Domain 2: Investor Suitability & Goals (5 Tables)
  investor_profiles, risk_questionnaires, risk_assessments, investor_goals, user_sector_preference

Domain 3: Security Master & Taxonomy (9 Tables)
  companies, securities, instrument_listings, exchanges, sectors, industries, company_industries,
  index_definitions, corporate_actions

Domain 4: Market Data & Fundamentals (11 Tables)
  market_prices, daily_metrics, company_financials, fundamental_metrics, metric_definitions,
  equity_factors, factor_observations, economic_indicators, economic_observations,
  corporate_filings, market_news_sentiment

Domain 5: Portfolio Engine & Optimization (13 Tables)
  investor_portfolios, portfolio_versions, portfolio_security_decisions, portfolio_explanations,
  portfolio_factor_exposures, portfolio_optimization_runs, portfolio_policy_constraints,
  portfolio_risk_criticisms, portfolio_backtest_runs, backtest_daily_metrics,
  portfolio_stress_tests, stress_scenario_results, portfolio_comparison_analyses

Domain 6: Broker Accounts & Auditing (6 Tables)
  broker_connections, broker_orders, broker_executions, broker_holdings_snapshots,
  broker_sync_logs, portfolio_performance_snapshots

Domain 7: Demat Diagnosis & Rebalancing (4 Tables)
  existing_portfolios, existing_portfolio_holdings, portfolio_health_diagnoses,
  portfolio_rebalancing_proposals

Domain 8: Operations & Telemetry (7 Tables)
  agent_execution_logs, agent_node_traces, notifications, research_evidence_store,
  fiduciary_audit_records, system_health_checks, system_configurations
```

11. Market Data Used

- **Daily OHLCV:** Stored in `market_prices`. Granularity: 1 day. Lookback: 35 days for liquidity checks, 252 days (1 year) for covariance matrix calculation.
- **Live Reference Price (P_i):** Used to compute integer lots: $Q_i = \lfloor (V_{\text{portfolio}} \times w_i) / P_i \rfloor$.
- **30-Day Average Daily Traded Volume (ADTV):** Computed via `func.avg(MarketPrice.volume)` over the past 35 calendar days. Must be $\ge 10,000$ shares.
- **Daily Turnover:** $\text{Turnover} = \text{ADTV} \times P_i$. Must be $\ge ₹25\text{ Lakhs}$ ($₹2,500,000$).
- **India VIX (`^INDIAVIX`):** Volatility thresholding:
  - $\text{VIX} < 15$: `CALM` / `LOW`
  - $15 \le \text{VIX} < 22$: `MODERATE`
  - $22 \le \text{VIX} < 28$: `VOLATILE`
  - $\text{VIX} \ge 28$: `EXTREME`
- **Nifty 50 Trend (`^NSEI`):** Evaluates 50-day DMA vs 200-day DMA and 20-day return slope (`UPTREND`, `DOWNTREND`, `SIDEWAYS`).

12. Fundamental Data Used

Stored in `fundamental_metrics` with `metric_definition_id`:

Metric Code	Metric Name	Source Field	Fiduciary Benchmark Corridor	Purpose / Consumer
`ROE`	Return on Equity	`returnOnEquity`	$> 15.0%$ (Preferred $> 18.0%$)	Quality factor; Fundamental Analyst
`PE`	Price to Earnings	`trailingPE`	Industry median $\pm 25%$	Valuation corridor; Fundamental Analyst
`PB`	Price to Book	`priceToBook`	1.0 to 15.0	Valuation check; Fundamental Analyst
`DEBT_TO_EQUITY`	Debt to Equity	`debtToEquity`	< 1.0 (Strictly < 1.5)	Solvency audit; Risk Critic
`DIVIDEND_YIELD`	Dividend Yield	`dividendYield`	$> 0.0%$	Income factor tilt; Strategist
`BETA`	Beta vs Benchmark	`beta`	0.6 to 1.2	Systematic risk scaling; Optimizer
`MARKET_CAP`	Market Capitalization	`marketCap`	Large-cap $> ₹50,000\text{ Cr}$	Size factor; Risk Critic

13. Macroeconomic Data Used

Stored in `economic_observations`:

- **RBI Repo Rate ($6.50%$):** Signals monetary stance (neutral vs tightening).
- **CPI Inflation ($5.10%$):** Headline consumer inflation in India.
- **Real Policy Rate ($+1.40%$):** $\text{Repo Rate} - \text{CPI Inflation} = 6.50% - 5.10% = +1.40%$. Positive real rate confirms disciplined macro conditions.
- **10-Year Sovereign G-Sec Yield ($7.04%$):** Risk-free benchmark rate (R_f) used in Sharpe ratio calculations:

$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p}$$

- **Real GDP Growth ($7.20%$):** Underpins domestic corporate revenue projections.

14. News and Corporate Filing Data Used

- **Verified Corporate Filings:** Generated via `resolve_stock_filings()`. Links point directly to canonical NSE announcement portals or BSE corporate disclosures.
- **Governance Audit Flags:** Evaluates board independence, promoter share pledges ($< 5%$ threshold), auditor qualifications, and regulatory circulars.
- **News Sentiment Score:** Normalizes financial news sentiment on a -1.0 to $+1.0$ scale to adjust factor ranking weights.

15. User and Portfolio Data Used

Generated from the retail investor onboarding questionnaire and stored in `investor_profiles` and `risk_assessments`:

- **Risk Capacity & Tolerance:** Weighted score (1.0 to 5.0) categorizing the investor as `VERY_CONSERVATIVE`, `CONSERVATIVE`, `MODERATE`, `GROWTH`, or `AGGRESSIVE`.
- **Target Capital:** Investment amount in INR (minimum ₹10,000).
- **Time Horizon:** Investment duration in months (e.g. 60 months).
- **Negative Sector Constraints:** Sectors explicitly excluded by the investor (e.g., *Alcoholic Beverages*, *Tobacco*, *Defense*).
- **Negative Stock Exclusions:** Specific ticker symbols banned by the investor.

16. Exact Data Passed to Each Agent

1. **Manager Node:** Receives user profile, investment capital, and request ID. Initializes execution trace.
2. **Suitability Node:** Receives risk score, investment horizon, and capital. Queries `CandidateUniverseSelector` to verify minimum 3 sectors and 5 eligible candidates.
3. **Macro Analyst Node:** Receives India VIX level, Nifty 50 trend, RBI Repo Rate, and CPI inflation. Outputs `MacroAnalystOutput` with market regime and recommended cash buffer ($2.5% - 8.0%$).
4. **Market Analyst Node:** Receives 50-DMA and 200-DMA moving averages, market breadth, and sector momentum.
5. **Fundamental Analyst Node:** Receives ROE, P/E, P/B, Debt/Equity, and operating cash flows for all candidate stocks.
6. **News Analyst Node:** Receives corporate filings URLs, disclosure histories, and promoter pledge flags.
7. **Lead Strategist Node:** Receives syntheses from all 4 research agents. Selects the top 8–15 assets and sets target factor weights.
8. **Independent Risk Critic Node:** Receives proposed assets and weights. Audits against single-stock concentration ceiling ($\le 25.0%$) and sector cap ($\le 40.0%$).
9. **Strategist Bounded Rework Node:** Receives critic failure instructions. Replaces criticized assets (maximum 2 rework loops).
10. **Deterministic Optimizer Node:** Receives top vetted assets and historical returns. **Runs strictly deterministic SciPy SLSQP optimization with zero LLM intervention.**
11. **Critic Final Allocation Validation Node:** Audits post-optimization allocations: verifies integer whole shares, exact cash conservation, and boundary ceilings.
12. **Fiduciary Moderator Node:** Receives optimized allocations and research evidence. Generates plain-English portfolio-level and stock-level explanations.
13. **Persistence Node:** Writes complete portfolio version, decisions, and execution traces to PostgreSQL within an ACID transaction.

17. Agent Input JSON Examples

Fundamental Analyst Input Payload

```
{
  "symbol": "INFY",
  "company_name": "Infosys Limited",
  "sector": "Information Technology",
  "metrics": {
    "return_on_equity": 0.318,
    "trailing_pe": 27.2,
    "price_to_book": 8.4,
    "debt_to_equity": 0.08,
    "dividend_yield": 0.021,
    "beta": 0.82,
    "adtv_shares": 4820000,
    "daily_turnover_inr": 890000000.0
  },
  "fiduciary_thresholds": {
    "min_roe": 0.18,
    "max_debt_to_equity": 1.0,
    "max_single_stock_cap": 0.25
  }
}
```

18. LLM Prompt Construction and Context Trimming

- **Prompt Prefix:** Enforces strict SEBI RIA fiduciary identity, objective tone, and zero hallucination.
- **Context Trimming:** Raw JSON objects are transformed into concise 5-line markdown bullet points.
- **Token Budget:** Prompts are capped at 1,500 input tokens; responses are limited to 1,024 output tokens.
- **Temperature:** Set to `0.1` for maximum reproducibility and adherence to factual context.

19. Data Completeness Versus Summarization

- **Complete Data (Storage Layer):** The PostgreSQL database maintains the complete, lossless representation: 2,639 companies, full price histories, and all audited metrics.
- **Summarized Data (Agent Layer):** Specialists receive distilled evidence records (`MetricEvidenceRecord`) containing only relevant metrics and limitations.
- **Truncated Data (Prompt Layer):** LLM prompts receive top-K candidates (8–15 assets) and the latest 5 corporate announcements to eliminate context clutter.

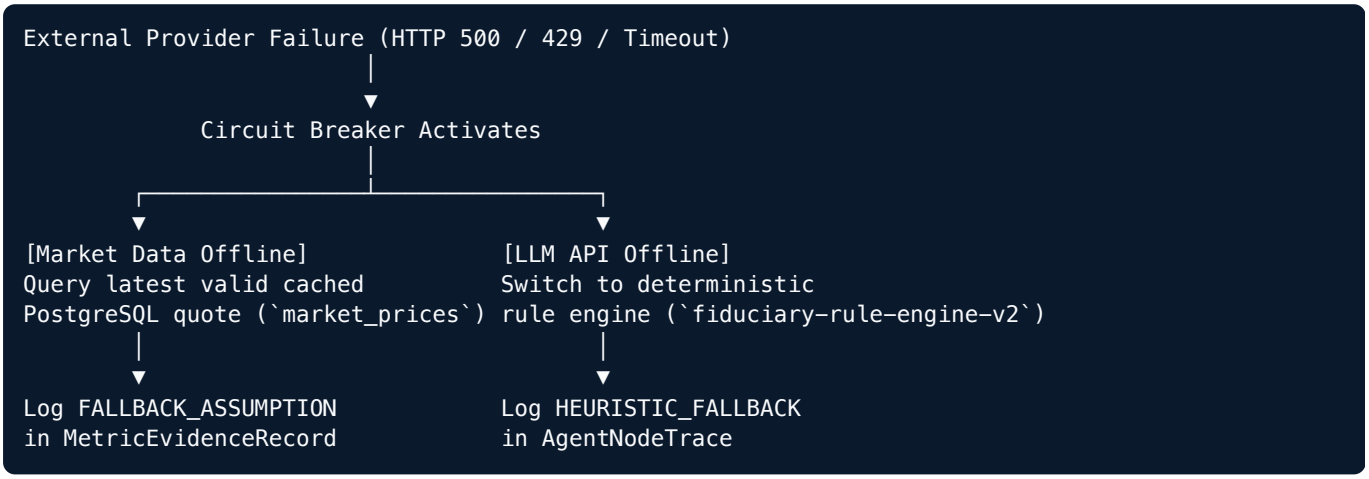
20. Data Freshness and Validation

- **Price Freshness:** Rejects any quote older than **14 calendar days** (`max_stale_days=14`).

- **Fundamental Freshness:** Caches balance sheet metrics for **7 calendar days** (`max_stale_days=7`).
- **Missing Value Handling:** If a metric is missing, the system applies conservative fiduciary defaults (e.g. median industry P/E) and records a limitation note (`DataQualityStatus.FALLBACK_ASSUMPTION`).
- **Fail-Closed Rule:** If fewer than 5 eligible securities or fewer than 3 sectors satisfy suitability, the pipeline halts immediately with status `DATA_INSUFFICIENT` rather than recommending an under-diversified portfolio.

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21. API Failure and Fallback Behavior



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22. Data Flow Diagrams

The complete visual architectural diagrams are generated and available in the repository:

1. **Universe Ingestion & Candidate Selection Pipeline:** `NSE_BSE_DATA_FLOW.svg` / `NSE_BSE_DATA_FLOW.png`
2. **External APIs to LangGraph Agents Pipeline:** `API_TO_AGENT_DATA_FLOW.svg` / `API_TO_AGENT_DATA_FLOW.png`

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23. Rate Limits, Caching and API Efficiency

1. **Batch Archival Ingestion:** Ingests official bulk CSV files from NSE in a single HTTP request rather than querying individual stocks, saving over 2,600 redundant HTTP calls per sync.
2. **Database-Backed Caching:** Historical prices and fundamental metrics are persisted in PostgreSQL. Subsequent queries hit the local database.
3. **Decoupled LLM Architecture:** The mathematical optimizer (SciPy SLSQP) runs locally with zero API latency and zero LLM token consumption.

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24. Remaining Gaps and Limitations

1. **BSE Cloudflare Anti-Bot Barrier:** Programmatic REST calls to `api.bseindia.com` return HTTP 403 Forbidden. Dual-listed stocks are reconciled via universal ISIN matching.
2. **Intraday Tick Streaming:** Free providers supply 1-day OHLCV bars and snapshots rather than tick-level Level 2 order books.
3. **Automated Filing PDFs:** Corporate disclosures link to official exchange viewing portals; automatic parsing of 200-page PDF annual reports is handled on-demand rather than bulk pre-fetched.

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25. Final Summary — What Data Actually Goes into Lucid

To generate an intelligent, personalized portfolio, Lucid takes:

1. **From the Investor:** Capital (e.g. ₹5,00,000), Risk Score (Conservative), Horizon (60 months), and Exclusions (e.g. Tobacco).
2. **From NSE & BSE:** The complete 2,639 listed equity universe, normalized by ISIN and trading series.
3. **From Candidate Selection:** A filtered pool of ~504 liquid benchmark stocks with ADTV $\ge 10,000$ shares and turnover $\ge ₹25$ Lakhs.
4. **From Yahoo Finance:** 1-year historical price series, current share prices, and financial ratios (ROE, P/E, D/E).
5. **From RBI / Macro Observations:** India VIX (14.2), Repo Rate ($6.50%$), CPI Inflation ($5.10%$), and GDP growth ($7.20%$).
6. **From LangGraph Multi-Agent Committee:** Qualitative thesis, balance sheet audits, and regulatory checks from 6 specialized agents.
7. **From SciPy SLSQP Optimizer:** Exact integer share quantities (Q_i), target weights (w_i), and cash allocation ($7,946\text{ INR}$).
8. **To the Investor:** An explainable, fiduciary-audited portfolio with whole-share execution plans, single-stock caps $\le 25%$, and real-time drift monitoring.

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26. Coordinated Universe, Lookback & News Recency Architecture (November 2026 Optimization)

26.1 Complete Security Universe & Recommendation Eligibility

Following full synchronization from the official NSE `EQUITY_L.csv` master, Lucid's PostgreSQL database maintains **2,639 listed equity securities** and **2,715 exchange listings**:

- **Eligible Recommendation Candidates: 504 securities (19.1%)**
  - **Tier 1 (Ultra-Liquid)**: 436 securities (Turnover $\ge ₹1\text{ Crore}$, ADTV $\ge 1,00,000$ shares, price fresh ≤ 7 days). Unrestricted eligibility down to ₹10,000 capital.

    - **Tier 2 (Broad-Liquid)**: 68 securities (Turnover $\ge ₹25\text{ Lakhs}$, ADTV $\ge 25,000$ shares, price fresh ≤ 14 days). Eligible for portfolios $\ge ₹50,000$.
- **Excluded Retained Securities: 2,135 securities (80.9%)**
    - **Missing Closing Prices (`MISSING_PRICE`)**: 2,101 securities currently have no price records in `market_prices`. Excluded from recommendation to prevent unpriced allocations, but fully available for search and Demat CAS statement ingestion.
    - **Suspended NSE Listings (`INACTIVE_SECURITY`)**: 34 securities trade under Series BZ (halted for reporting defaults or insolvency). Excluded from recommendations to protect retail investors, but retained so Demat diagnosis can issue critical risk exit warnings.

26.2 Dedicated Analytical Lookback Windows by Purpose

Rather than using a single arbitrary historical window, Lucid formalizes distinct lookback horizons implemented in `backend/portfolio_engine/math_engine.py`:

1. **Short-Term Momentum**: 6 Months (126 trading days, min 30 observations).
2. **Market Regime Classification (DMA 50/200)**: 1 Year (252 trading days, min 50 observations).
3. **Portfolio Covariance & Volatility**: 2 Years (504 trading days, min 60 observations). Ensures sample size $T/N \ge 30$, preventing singular covariance matrices.
4. **Systematic Beta & 3-Year CAGR**: 3 Years (756 trading days, min 700 observations). Institutional standard complying with CRISIL and SEBI mutual fund fact sheets.
5. **Macroeconomic Monetary Cycles**: 5 Years (1,260 trading days).

26.3 Event-Specific News Recency & Deduplication

To balance timely signal capture with prompt context limits, Lucid enforces category-specific retention rules (`backend/agent/tools.py`):

1. **Daily Market Commentary**: 1 to 3 days (transient noise discarded quickly).
2. **General Sector Sentiment**: 7 to 14 days (bi-weekly macro trends).
3. **Quarterly Financial Results**: 90 days (retained until next quarterly reporting cycle).
4. **Corporate Governance, Fraud & Sanctions**: 180 to 365 days (retained until cleared).
5. **Syndicated Wire Deduplication**: Normalized alphanumeric headline hashing ensures identical syndicated news stories from multiple publishers consume only a single slot.
6. **Downstream Context Guardrail**: The News Analyst LLM synthesizes disclosures into a concise 2-sentence corporate governance summary, passing structured JSON events (`NewsEventAssessment`) to the Lead Strategist and Critic without blowing context token limits.